

Redox Reactions

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

Q1 JEE Main 2020 - 2 September (Morning)

While titrating dilute HCl solution with aqueous NaOH, which of the following will not be required?

- (A) Pipette and distilled water
- (B) Clamp and phenolphthalein
- (C) Burette and porcelain tile
- (D) Bunsen burner and measuring cylinder

Q2 JEE Main 2020 - 3 September (Evening)

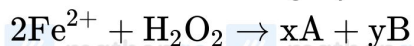
The volume (in mL) of 0.1 N NaOH required to neutralise 10mL of 0.1N phosphinic acid is

Q3 JEE Main 2020 - 4 September (Morning)

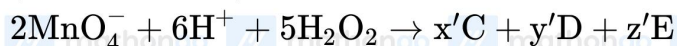
A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316g of KMnO_4 in acid solution. The purity of H_2O_2 (in %) is (mol. wt. of $\text{H}_2\text{O}_2 = 34$; mol. wt. of $\text{KMnO}_4 = 158$)

Q4 JEE Main 2020 - 4 September (Evening)

Consider the following equations:



(in basic medium)



(in acidic medium)

The sum of the stoichiometric coefficients

x, y, x', y' and z' for products A, B, C, D and E, respectively, is

Q5 JEE Main 2020 - 5 September (Evening)

The volume, in mL, of 0.02M $\text{K}_2\text{Cr}_2\text{O}_7$ solution required to react with 0.288g of ferrous oxalate in acidic medium is

(Molar mass of $\text{Fe} = 56\text{g mol}^{-1}$)

Q6 JEE Main 2020 - 6 September (Morning)

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Chemistry

Potassium chlorate is prepared by the electrolysis of KCl in basic solution

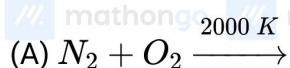


If only 60% of the current is utilized in the reaction, the time (rounded to the nearest hour) required to produce 10g of KClO_3 using a current of 2A is

(Given : $F = 96,500 \text{Cmol}^{-1}$; molar mass of $\text{KClO}_3 = 122 \text{gmol}^{-1}$)

Q7 JEE Main 2020 - 7 January (Evening)

Amongst the following which is redox reaction ?

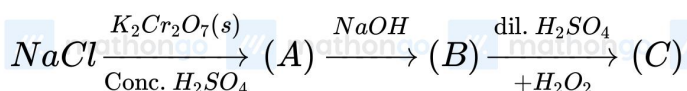


(B) Formation of O_3 from O_2

(C) Reaction between NaOH and H_2SO_4

(D) Reaction between AgNO_3 and NaCl

Q8 JEE Main 2020 - 7 January (Evening)



Determine total number of atoms in per unit formula of (A), (B) & (C)

Q9 JEE Main 2020 - 8 January (Morning)

The strength of an aqueous NaOH solution is most accurately determined by titrating :

(Note : consider that an appropriate indicator is used)

(A) NaOH in burette and oxalic acid in conical flask

(B) NaOH in burette and concentrated H_2SO_4 in conical flask

(C) NaOH in volumetric flask and concentrated H_2SO_4 in conical flask

(D) Oxalic acid in burette and NaOH in conical flask

Q10 JEE Main 2020 - 9 January (Morning)

The compound that cannot act both as oxidising and reducing agent is :

(A) H_2SO_3

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Chemistry

(B) HNO_2

(C) H_3PO_4

(D) H_2O_2

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Chemistry

Answer Key

Q1 (D)

Q2 (10)

Q3 (85)

Q4 (19)

Q5 (50)

Q6 (11)

Q7 (A)

Q8 (18)

Q9 (B)

Q10 (C)

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Hints & Solutions

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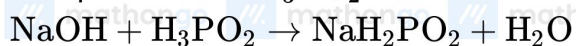
Chemistry

Q1

Bunsen Burner and measuring cylinder is not required for titration.

Q2

Phosphinic acid is H_3PO_2

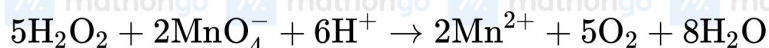


milliequivalents of NaOH = milliequivalents of H_3PO_2

$$V \times 0.1 = 10 \times 0.1$$

$$V = 10\text{mL}$$

Q3



$$\text{Moles of KMnO}_4 = \frac{0.316}{158} = 2 \times 10^{-3}$$

Equivalents of H_2O_2 = Equivalent of KMnO_4

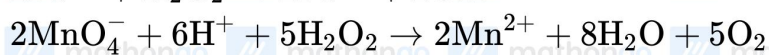
$$= 2 \times 10^{-3} \times 5 = 0.01$$

$$\text{Moles of H}_2\text{O}_2 = \frac{0.01}{2} = 0.005$$

$$\text{Mass of pure H}_2\text{O}_2 = 0.005 \times 34 = 0.170\text{gm}$$

$$\text{Percentage purity} = \frac{0.17}{0.2} \times 100 = 85\%$$

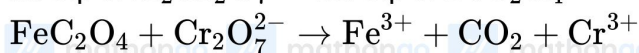
Q4



$$\therefore x + y + x' + y' + z' = 19$$

Q5

m. eq. of $\text{K}_2\text{Cr}_2\text{O}_7$ = m. eq. of FeC_2O_4



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Chemistry

$$\Rightarrow V \times 0.02 \times 6 = \frac{0.288 \times 3 \times 1000}{144}$$

$$V = 50 \text{ mL}$$

Q6



For synthesis of 1 mole of ClO_3^- , 6F of charge is required

current efficiency = 60%

$\therefore$ To synthesise 1 mole of ClO_3^- , 10F of charge is required.

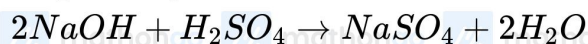
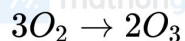
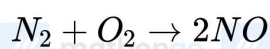
To synthesise $\frac{10}{122}$ moles of KClO_3 ,

$$\text{charge} = \frac{10 \times 10}{122} \text{ F}$$

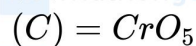
$$\therefore \frac{2 \times t(\text{hr}) \times 3600}{96500} = \frac{100}{122}$$

$$t = 11$$

Q7



Q8



Q9

Oxalic acid is a primary standard solution while H_2SO_4 is a secondary standard solution.

Q10

As in H_3PO_4 Phosphorous is present it's maximum oxidation number state hence it cannot act as reducing agent.

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Chemistry

